BRING-UP REPORT (SIMULATED BENCH)

PCB-0018-A_bldc-motor-driver

Run 20260930-194840-sim: passed, 103 of 103 steps passed.

103	0	0	41
steps passed	steps failed	steps not reached	human confirmations

Run

ITEM	VALUE
Board	PCB-0018-A_bldc-motor-driver
Bench	sim
Started	2026-09-30T19:48:40Z
Finished	2026-09-30T19:48:43Z
Procedure	6409f9a842f16f61 (generated 2026-09-30T19:48:24Z)
Firmware	360ed127fec7b60c
netlist	31d0f8db9afdf43f
constraints	0780106a0c9dc221
bom	f914bf2c56105f4f
requirements	acc5ff8498c62447
manifest	197c37c8064d88b9
Simulated faults	none
Seed	1

Stages

STAGE	STEPS	PASS	FAIL	NOT RUN
visual	2	2	0	0
unpowered	31	31	0	0
power-up	7	7	0	0
rails	14	14	0	0
programming	5	5	0	0
blocks	33	33	0	0
limits	11	11	0	0

Not in this procedure: full-function (firmware has no spin command yet (manifest stage sixstep)); limits:ov (over-voltage trip at 30 V is above the rated input max 28 V; not applied).

Measurements

STEP	WHAT	LIMITS	VALUE	VERDICT
unpowered.03	VM_IN to GND is not a short	>= 1000 ohm	7015.46 ohm	pass
unpowered.05	VM to GND is not a short	>= 1000 ohm	17711.5 ohm	pass
unpowered.07	+12V to GND is not a short	>= 100 ohm	1645.66 ohm	pass
unpowered.09	+5V to GND is not a short	>= 100 ohm	882.604 ohm	pass
unpowered.11	+3V3 to GND is not a short	>= 100 ohm	1243.15 ohm	pass
unpowered.13	VDDA to GND is not a short	>= 100 ohm	1174.24 ohm	pass
unpowered.15	PHASE_A to GND is not a short	>= 1000 ohm	14773.9 ohm	pass
unpowered.17	PHASE_B to GND is not a short	>= 1000 ohm	16830.9 ohm	pass
unpowered.19	PHASE_C to GND is not a short	>= 1000 ohm	6407.89 ohm	pass
unpowered.21	Q401 body diode conducts one way	0.3 .. 0.8 V	0.443402 V	pass
unpowered.23	Q402 body diode conducts one way	0.3 .. 0.8 V	0.540292 V	pass
unpowered.25	Q403 body diode conducts one way	0.3 .. 0.8 V	0.491932 V	pass
unpowered.27	Q404 body diode conducts one way	0.3 .. 0.8 V	0.531474 V	pass
unpowered.29	Q405 body diode conducts one way	0.3 .. 0.8 V	0.440253 V	pass

STEP	WHAT	LIMITS	VALUE	VERDICT
unpowered.31	Q406 body diode conducts one way	0.3 .. 0.8 V	0.493446 V	pass
power-up.06	idle input current	0.002 .. 0.18 A	0.0347532 A	pass
rails.02	VM voltage	11.4 .. 12.12 V	11.8475 V	pass
rails.04	+12V voltage	11.04 .. 12.96 V	12.2565 V	pass
rails.06	+5V voltage	4.75 .. 5.25 V	5.06021 V	pass
rails.08	+3V3 voltage	3.135 .. 3.465 V	3.25353 V	pass
rails.10	VDDA voltage	3.135 .. 3.465 V	3.25302 V	pass
rails.12	+12V ripple	<= 0.24 V	0.0629817 V	pass
rails.14	+5V ripple	<= 0.1 V	0.0381745 V	pass
programming.02	flash			pass
programming.04	boot banner		fwe PCB-0018-A sim	pass
programming.05	firmware answers		OK {”board”:”PCB-0018-A”}	pass
blocks.01	firmware test hook version		OK {”board”:”PCB-0018-A”}	pass
blocks.02	firmware test hook status		OK {”sim”:true,”armed”:false,”vbus_v”:11.971}	pass
blocks.03	firmware test hook selftest		OK {”sim”:true}	pass
blocks.04	firmware test hook adc		OK {”sim”:true}	pass
blocks.05	firmware reads the input voltage		vbus_v=11.932	pass
blocks.07	U501 sense-amp offset	1.6 .. 1.7 V	1.64766 V	pass
blocks.09	U502 sense-amp offset	1.6 .. 1.7 V	1.63587 V	pass
blocks.11	U503 sense-amp offset	1.6 .. 1.7 V	1.64165 V	pass
blocks.14	SW601 seen by firmware		fwe PCB-0018-A sim	pass
blocks.16	SW602 seen by firmware		EVT {”boot”:”power”}	pass
blocks.18	Hall inputs toggle	>= 2 edges	HALL_A=12, HALL_B=12, HALL_C=12	pass
blocks.20	reply to arm		OK {”sim”:true}	pass
blocks.21	reply to duty 0.5 0 0		OK {”sim”:true}	pass

STEP	WHAT	LIMITS	VALUE	VERDICT
blocks.23	PHASE_A switches (Q401/Q402)	0.45 .. 0.55	0.498758	pass
blocks.24	PHASE_A PWM frequency	19600 .. 20400 Hz	19999 Hz	pass
blocks.25	reply to duty 0 0.5 0		OK {”sim”:true}	pass
blocks.27	PHASE_B switches (Q403/Q404)	0.45 .. 0.55	0.494662	pass
blocks.28	PHASE_B PWM frequency	19600 .. 20400 Hz	19935.4 Hz	pass
blocks.29	reply to duty 0 0 0.5		OK {”sim”:true}	pass
blocks.31	PHASE_C switches (Q405/Q406)	0.45 .. 0.55	0.494376	pass
blocks.32	PHASE_C PWM frequency	19600 .. 20400 Hz	19990.3 Hz	pass
blocks.33	reply to disarm		OK {”sim”:true}	pass
limits.03	idle current at 10 V	0.002 .. 0.18 A	0.0391591 A	pass
limits.06	idle current at 28 V	0.002 .. 0.18 A	0.0160505 A	pass
limits.08	UV trip below 9 V		EVT {”fault”:”uv”,”vbus_v”:8.50}	pass
limits.10	reply to clear		OK {”sim”:true}	pass

STEP	ACTION	BY	AT	RESULT

Human steps

A simulated run confirms its own human steps as `sim`.

STEP	ACTION	BY	AT	RESULT
visual.01	Check the board for damage, solder bridges on fine-pitch ICs (U101, U201, U202, U203, U301, U501, U502, U503, U601) and missing parts against fab/BOM.csv.	sim	2026-09-30T19:48:40Z	pass
visual.02	Check orientation of polarised parts: D101 SMCJ28A, D201 MBR0520L, D601 LED green (power), D602 LED yellow (status), D603 LED red (fault), D701 PESD5V0S1BA-ES, D702 PESD5V0S1BA-ES, D703 PESD5V0S1BA-ES, D704 PESD5V0S1BA-ES, D705 PESD5V0S1BA-ES, D709 PESD5V0S1BA-ES, D710 PESD5V0S1BA-ES, D711 PESD5V0S1BA-ES, U101 LM74500-Q1, U201 LM5164, U202 TPS61040, U203 TLV75733, U301 DRV8300DRGER, U501 INA240A1, U502 INA240A1, U503 INA240A1, U601 STM32G431CBT6, Q101 NTMFS5C628NL, Q401 NTMFS5C628NL, Q402 NTMFS5C628NL, Q403 NTMFS5C628NL, Q404 NTMFS5C628NL, Q405 NTMFS5C628NL, Q406 NTMFS5C628NL, C101 470uF 63V, C104 470uF 63V.	sim	2026-09-30T19:48:40Z	pass
unpowered.01	Board unpowered, every cable off. Discharge the input capacitors.	sim	2026-09-30T19:48:40Z	pass
unpowered.02	DMM on resistance: red (+) on TP101 (VM_IN), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.04	DMM on resistance: red (+) on C101 pin 1 (VM), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.06	DMM on resistance: red (+) on C206 pin 1 (+12V), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass

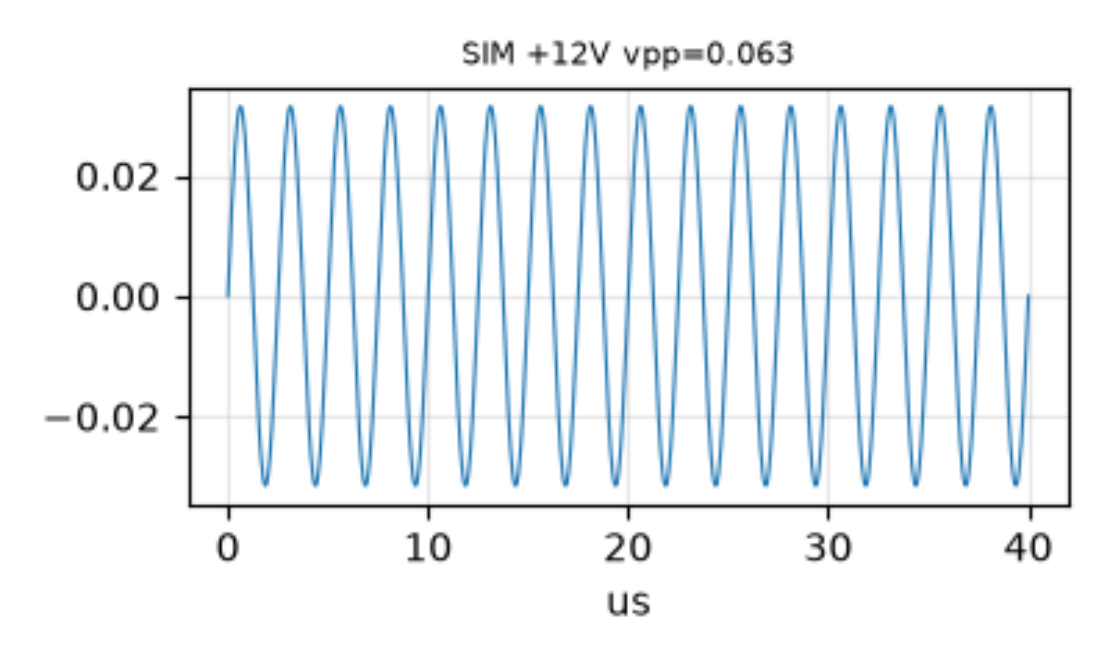
STEP	ACTION	BY	AT	RESULT
unpowered.08	DMM on resistance: red (+) on C202 pin 1 (+5V), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.10	DMM on resistance: red (+) on C208 pin 1 (+3V3), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.12	DMM on resistance: red (+) on C209 pin 1 (VDDA), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.14	DMM on resistance: red (+) on TP401 (PHASE_A), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.16	DMM on resistance: red (+) on TP402 (PHASE_B), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.18	DMM on resistance: red (+) on TP403 (PHASE_C), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
unpowered.20	DMM on diode: red (+) on TP401 (PHASE_A), black (-) on C101 pin 1 (VM).	sim	2026-09-30T19:48:40Z	pass
unpowered.22	DMM on diode: red (+) on C432 pin 2 (LS_SRC_A), black (-) on TP401 (PHASE_A).	sim	2026-09-30T19:48:40Z	pass
unpowered.24	DMM on diode: red (+) on TP402 (PHASE_B), black (-) on C101 pin 1 (VM).	sim	2026-09-30T19:48:40Z	pass
unpowered.26	DMM on diode: red (+) on C434 pin 2 (LS_SRC_B), black (-) on TP402 (PHASE_B).	sim	2026-09-30T19:48:40Z	pass
unpowered.28	DMM on diode: red (+) on TP403 (PHASE_C), black (-) on C101 pin 1 (VM).	sim	2026-09-30T19:48:40Z	pass
unpowered.30	DMM on diode: red (+) on C436 pin 2 (LS_SRC_C), black (-) on TP403 (PHASE_C).	sim	2026-09-30T19:48:40Z	pass
power-up.02	Connect the supply: + to TP101 (VM_IN), - to TP102 (GND). Supply set to 12 V / 0.2 A, output still off.	sim	2026-09-30T19:48:40Z	pass
power-up.03	About to switch the supply on at 12 V with a 0.2 A limit. Hands clear; watch for smoke or a hot part.	sim	2026-09-30T19:48:40Z	pass
power-up.07	Touch-test: no part is hot a few seconds after power-up.	sim	2026-09-30T19:48:40Z	pass
rails.01	DMM on DC volts: red (+) on C101 pin 1 (VM), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
rails.03	DMM on DC volts: red (+) on C206 pin 1 (+12V), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass

STEP	ACTION	BY	AT	RESULT
rails.05	DMM on DC volts: red (+) on C202 pin 1 (+5V), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
rails.07	DMM on DC volts: red (+) on C208 pin 1 (+3V3), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
rails.09	DMM on DC volts: red (+) on C209 pin 1 (VDDA), black (-) on TP102 (GND).	sim	2026-09-30T19:48:40Z	pass
rails.11	Scope channel 1 on C206 pin 1 (+12V), ground spring to the nearest GND pad, AC coupled, 20 MHz bandwidth limit.	sim	2026-09-30T19:48:40Z	pass
rails.13	Scope channel 1 on C202 pin 1 (+5V), ground spring to the nearest GND pad, AC coupled, 20 MHz bandwidth limit.	sim	2026-09-30T19:48:43Z	pass
programming.01	Connect the SWD probe to J601; the board stays powered from the supply.	sim	2026-09-30T19:48:43Z	pass
programming.03	Connect the USB-UART (3V3, 115200 8N1) to J701: adapter RX to pin 3 (board TX), adapter TX to pin 4 (board RX), GND to GND. Then press reset.	sim	2026-09-30T19:48:43Z	pass
blocks.06	DMM on DC volts: red (+) on R501 pin 1 (CSA_A_OUT), black (-) on TP102 (GND).	sim	2026-09-30T19:48:43Z	pass
blocks.08	DMM on DC volts: red (+) on R502 pin 1 (CSA_B_OUT), black (-) on TP102 (GND).	sim	2026-09-30T19:48:43Z	pass
blocks.10	DMM on DC volts: red (+) on R503 pin 1 (CSA_C_OUT), black (-) on TP102 (GND).	sim	2026-09-30T19:48:43Z	pass
blocks.12	Note which LEDs are lit: D601 LED green (power), D602 LED yellow (status), D603 LED red (fault).	sim	2026-09-30T19:48:43Z	pass
blocks.13	Press and release SW601 (Reset switch SMD).	sim	2026-09-30T19:48:43Z	pass
blocks.15	Press and release SW602 (User switch SMD).	sim	2026-09-30T19:48:43Z	pass
blocks.17	Connect the logic analyser to the Hall inputs (HALL_A, HALL_B, HALL_C) at J702 pins 3, 4, 5; plug in the Hall sensor and turn the motor slowly by hand during the capture.	sim	2026-09-30T19:48:43Z	pass
blocks.19	Motor DISCONNECTED. The next steps switch the bridge with no load at 12 V. Confirm the phase outputs are free.	sim	2026-09-30T19:48:43Z	pass

STEP	ACTION	BY	AT	RESULT
blocks.22	Scope channel 1 on TP401 (PHASE_A), DC coupled, 10x probe.	sim	2026-09-30T19:48:43Z	pass
blocks.26	Scope channel 1 on TP402 (PHASE_B), DC coupled, 10x probe.	sim	2026-09-30T19:48:43Z	pass
blocks.30	Scope channel 1 on TP403 (PHASE_C), DC coupled, 10x probe.	sim	2026-09-30T19:48:43Z	pass

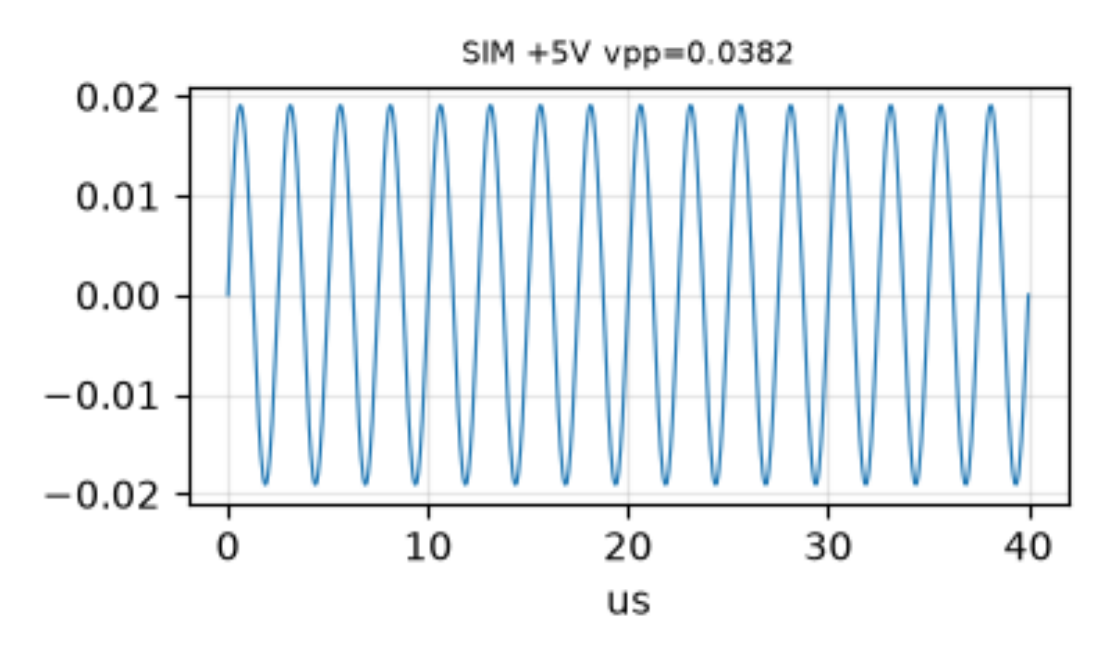
Scope captures

RAILS.12



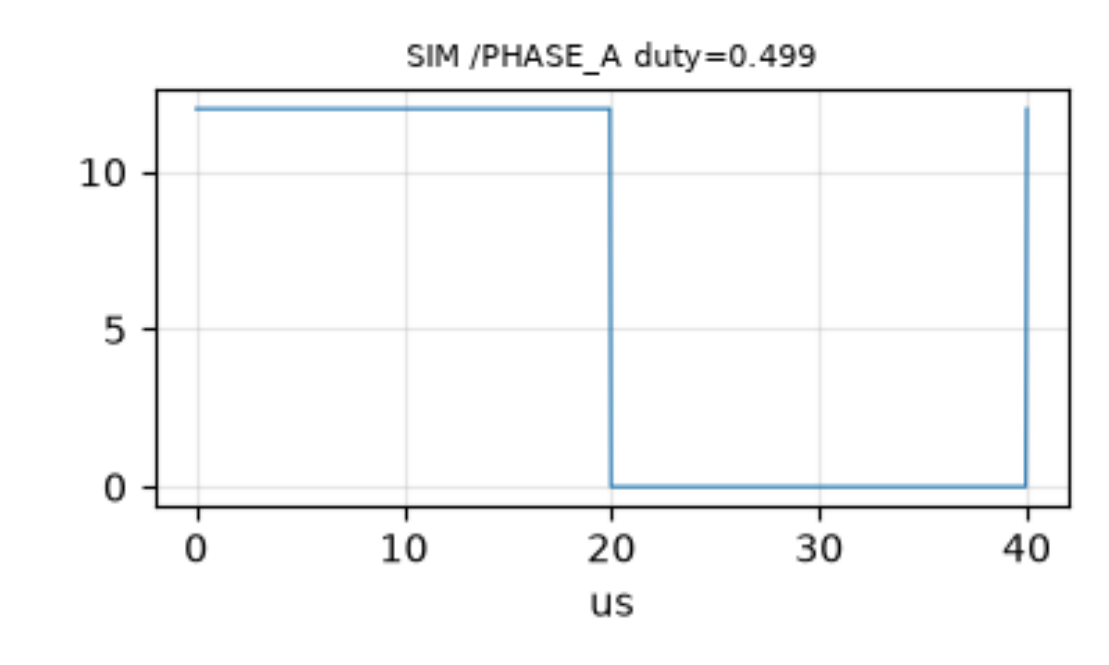
+12V ripple (captures/rails.12.png)

RAILS.14



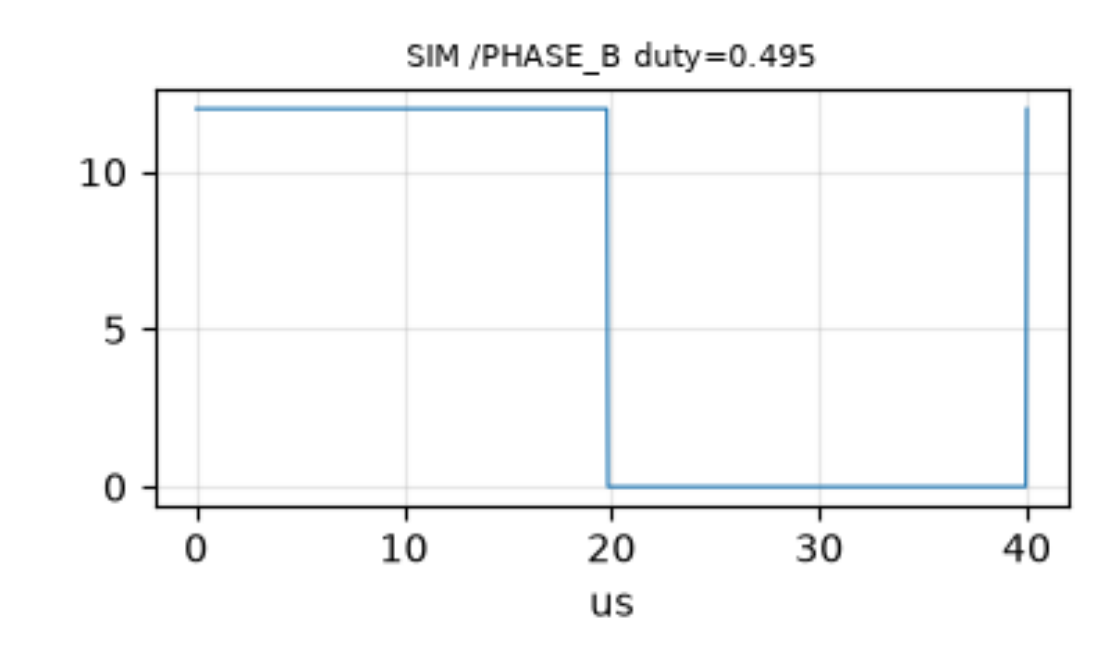
+5V ripple (captures/rails.14.png)

BLOCKS.23



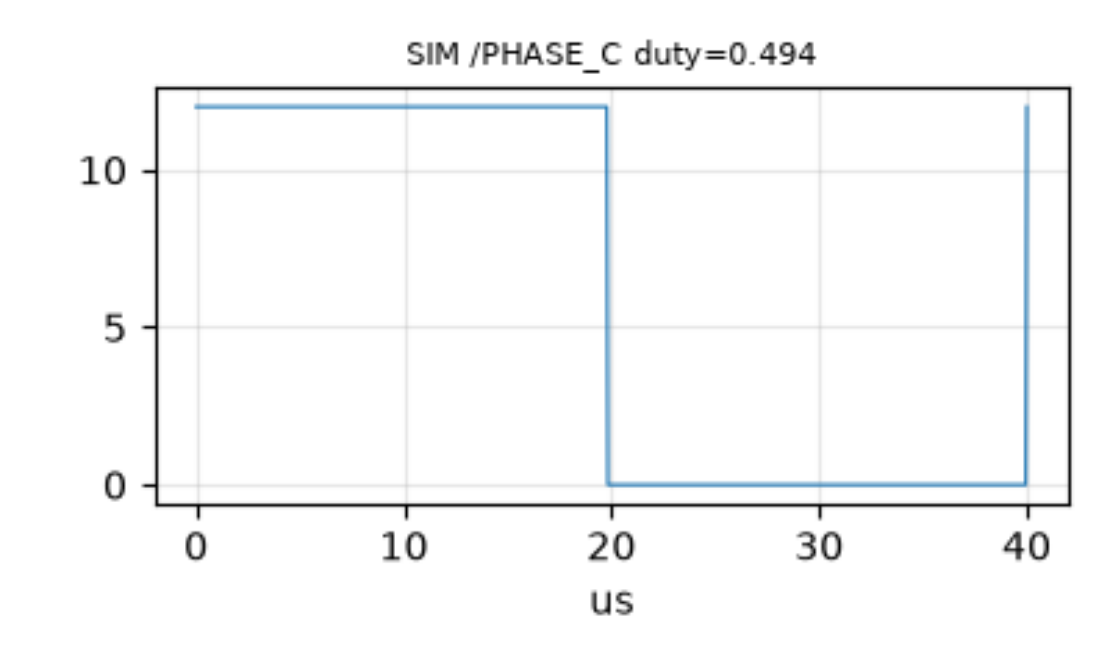
PHASE_A switches (Q401/Q402) (captures/blocks.23.png)

BLOCKS.27



PHASE_B switches (Q403/Q404) (captures/blocks.27.png)

BLOCKS.31



PHASE_C switches (Q405/Q406) (captures/blocks.31.png)